

MATH 4820 / Regression Analysis

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**Polynomial
Regression
Models**

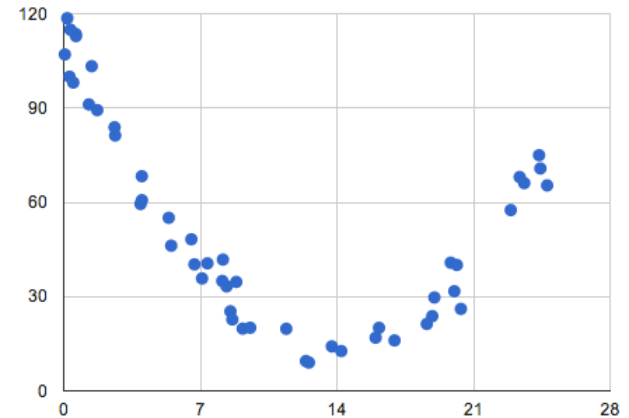


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7.1 INTRODUCTION

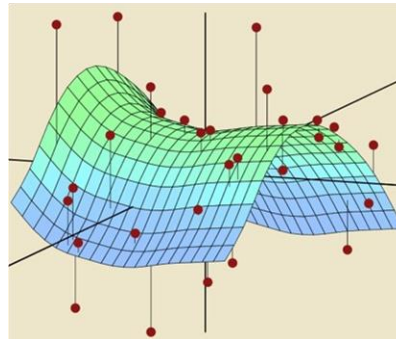
A second-order polynomial in one variable:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \varepsilon$$



A second-order polynomial in two variables:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{11} x_1^2 + \beta_{22} x_2^2 + \beta_{12} x_1 x_2 + \varepsilon$$



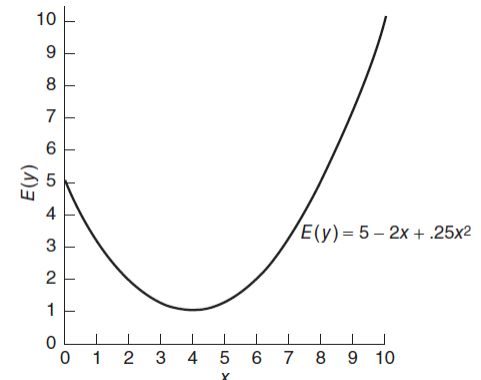
7.2 POLYNOMIAL MODELS IN ONE VARIABLE

- One-variable form, again:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \varepsilon$$

- If we let $x_1 = x$ and $x_2 = x^2$, we have the same type of model as in previous chapters – standard linear regression analysis applies.
- The expectation of y for a one-variable polynomial model is

$$E(y) = \beta_0 + \beta_1 x + \beta_2 x^2$$



7.2 POLYNOMIAL MODELS IN ONE VARIABLE

Cautions in fitting a polynomial in one-variable

1. Keep the order of the model as low as possible.
 - This is especially true if you are using the model as a predictor.
 - Transformations are often preferred over higher-order models
 - Parsimony – this is a good thing, try to fit the data using the simplest model possible.
 - Remember: You can always fit an $n - 1$ order model to a set of data with n points. But this is undesirable.

7.2 POLYNOMIAL MODELS IN ONE VARIABLE

Cautions in fitting a polynomial in one-variable

2. Model Building Strategy.

- One approach is fitting the lowest order polynomial possible and build up (forward selection).
- Second approach is fitting the highest order polynomial of interest, and removing terms (backward elimination).
- In general, you may not get the same result from the two approaches. You should always try to fit the lowest-order possible.

3. Extrapolation

Can be dangerous when the model is a higher-order polynomial.

The nature of the true underlying relationship may change or be completely different than the system that produced the data used to fit the model.

7.2 POLYNOMIAL MODELS IN ONE VARIABLE

Cautions in fitting a polynomial in one-variable

4. Ill-conditioning

- Ill-conditioning refers to the fact that as the order of the model increases, the $\mathbf{X}'\mathbf{X}$ matrix inversion will become inaccurate – error can be introduced into the parameter estimates.
- As the order of the model $\uparrow$, multicollinearity $\uparrow$
- Centering the variables first may remove some ill-conditioning but not all.

7.2 POLYNOMIAL MODELS IN ONE VARIABLE

Cautions in fitting a polynomial in one-variable

5. Hierarchy

- *hierarchical model* is one which, if it is of order n , then it contains all terms with orders of n and below:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \cdots + \beta_{n-1} x^{n-1} + \beta_n x^n \varepsilon$$

- Two schools of thought:
 - 1) Maintain hierarchy
 - 2) Maintaining hierarchy is not important.
- What to do? Fit the model with only significant terms and use knowledge and understanding of the process to determine if a hierarchical model is necessary (if you do not have one).

7.2 POLYNOMIAL MODELS IN ONE VARIABLE

Centering

Sometimes, centering the regressor variables can minimize or eliminate at least some of the ill-conditioning that may be present in a polynomial model.

Example 7.1 The Hardwood Data

TABLE 7.1 Hardwood Concentration in Pulp and Tensile Strength of Kraft Paper, Example 7.1

Hardwood Concentration, x_i (%)	Tensile Strength, (psi) y , (psi)
1	6.3
1.5	11.1
2	20.0
3	24.0
4	26.1
4.5	30.0
5	33.8
5.5	34.0
6	38.1
6.5	39.9
7	42.0
8	46.1
9	53.1
10	52.0
11	52.5
12	48.0
13	42.8
14	27.8
15	21.9

EXAMPLE 7.1 THE HARDWOOD DATA

CENTERING

The regression analysis is provided below.

The regression equation is
 $y = -6.67 + 11.8x - 0.635x^2$

Predictor	Coef	SE Coef	T	P	VIF
Constant	-6.674	3.400	-1.96	0.067	
x	11.764	1.003	11.73	0.000	17.1
x2	-0.63455	0.06179	-10.27	0.000	17.1

S = 4.420 R-Sq = 90.9% R-Sq(adj) = 89.7%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	2	3104.2	1552.1	79.43	0.000
Residual Error	16	312.6	19.5		
Total	18	3416.9			

Note that the variance inflation factors indicate that multicollinearity may be a problem.

EXAMPLE 7.1 THE HARDWOOD DATA

CENTERING

Now, center the data using the mean of the regressor variable. The new data is given as

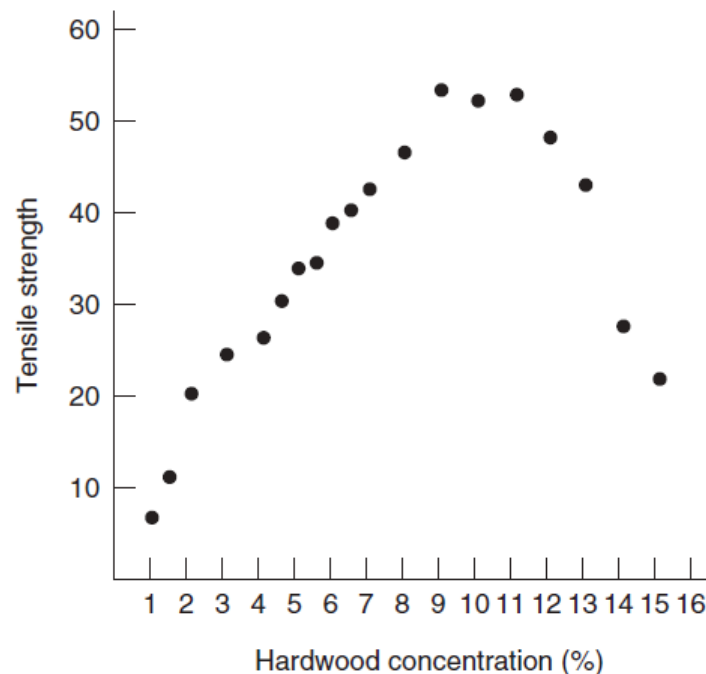


Figure 7.3 Scatterplot of data, Example 7.1.

$x_i - 7.2632$	$(x_i - 7.2632)^2$	y
-6.2632	39.2277	6.3
-5.7632	33.2145	11.1
-5.2632	27.7013	20.0
-4.2632	18.1749	24.0
-3.2632	10.6485	26.1
-2.7632	7.6353	30.0
-2.2632	5.1221	33.8
-1.7632	3.1089	34.0
-1.2632	1.5957	38.1
-0.7632	0.5825	39.9
-0.2632	0.0693	42.0
0.7368	0.5429	46.1
1.7368	3.0165	53.1
2.7368	7.4901	52.0
3.7368	13.9637	52.5
4.7368	22.4373	48.0
5.7368	32.9109	42.8
6.7368	45.3845	27.8
7.7368	59.8581	21.9

EXAMPLE 7.1 THE HARDWOOD DATA

CENTERING

$$y = \beta_0 + \beta_1(x - 7.2632) + \beta_2(x - 7.2632)^2 + \varepsilon$$

Now, a new model is fit:

The regression equation is
 $y = 45.3 + 2.55 \text{ xcent} - 0.635 \text{ x2cent}$

Predictor	Coef	SE Coef	T	P
Constant	45.295	1.483	30.55	0.000
xcent	2.5463	0.2538	10.03	0.000
x2cent	-0.63455	0.06179	-10.27	0.000

VIF
1.1
1.1

S = 4.420 R-Sq = 90.9% R-Sq(adj) = 89.7%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	2	3104.2	1552.1	79.43	0.000
Residual Error	16	312.6	19.5		
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QUESTIONS?

- ANY QUESTION?